CLOUD SECURITY

Hands-on Tasks Report

AWS environment exploration, incident analysis, cloud log review, misconfiguration assessment, and security recommendations.

Assessment scope	IAM configuration, security event evidence, AWS CloudTrail event history, and cloud security controls.
Environment	AWS Management Console and supplied security-monitoring screenshots.
Overall rating	**Medium Risk, Requires Remediation**

Prepared from the task notes and screenshots provided. Findings are limited to the visible evidence; suspicious activity is not confirmed malicious activity without further investigation.

Task 1: Explore Cloud Environments

Objective

Gain familiarity with cloud resources, identity management, and cloud service navigation.

Activities Performed

- Accessed the AWS Management Console.
- Opened Identity and Access Management (IAM).
- Reviewed the IAM user configuration, attached permissions, console-access status, and available security-credential controls.
- Examined the account's visible security-related settings.

Evidence: IAM User Review

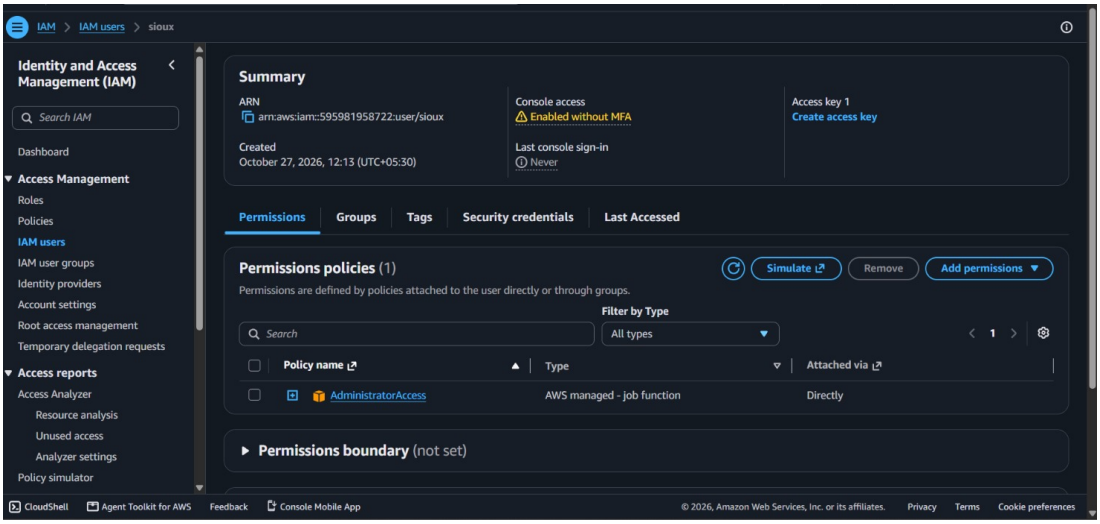


Figure 1. AWS IAM user summary showing an attached AdministratorAccess policy and console access enabled without MFA.

Observations

- The environment contains IAM users and policies that define access to AWS resources.
- Permissions can be attached directly to a user or inherited through groups. IAM roles provide another way to grant temporary permissions.
- The screenshot shows one user with the AWS-managed AdministratorAccess policy attached directly.

Findings

- **Privileged access:** AdministratorAccess is attached directly to the reviewed IAM user.
- **Console access:** Enabled.
- **MFA status:** The console summary displays “Enabled without MFA.”
- **Access key:** The page offers an option to create an access key; the screenshot does not establish that an active access key already exists.

Recommendation: Enable MFA for privileged accounts, apply least-privilege permissions, and prefer role-based access or IAM groups where appropriate. Create access keys only when required, protect them carefully, and rotate or disable unused keys.

Task 2: Analyze Cloud Security Incidents

Objective

Investigate suspicious activities and security events within the cloud environment.

Activities Performed

- Reviewed the supplied security-monitoring event table and timeline.
- Compared event times, device names, usernames, domain, and object names.
- Looked for repeated activity affecting a sensitive Windows registry path.

Evidence: Security Incident Timeline

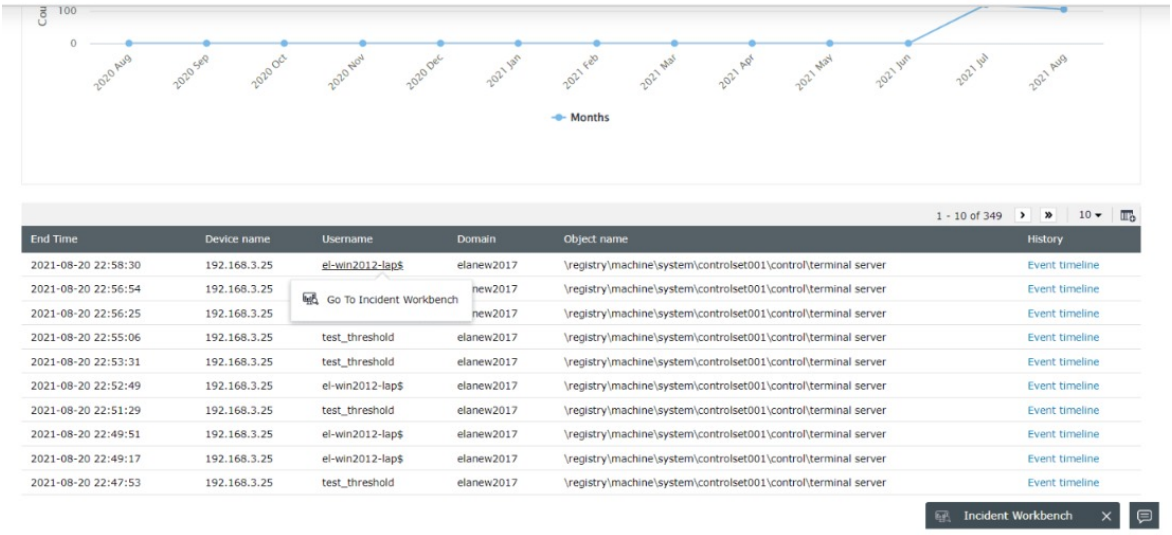


Figure 2. Supplied monitoring view showing repeated events associated with a terminal-server-related registry path.

Observations and Findings

- Multiple events appear close together in time and reference the same device address shown in the screenshot.
- The object name repeatedly references a registry path under the system control set and terminal server configuration.
- More than one username appears in the visible records.
- The screenshot shows repeated changes or event records, but does not by itself prove that the changes were unauthorized or malicious.

Potential Security Impact

- Unauthorized configuration changes or policy tampering.
- Misuse of legitimate user credentials or possible insider activity.
- Malware-related persistence or system manipulation, if corroborated by other evidence.

Recommendation: Preserve the relevant event records, investigate the affected host, validate whether the changes were approved, and correlate timestamps with authentication logs, endpoint detection alerts, process execution records, and change tickets. Escalate only when supporting evidence warrants it.

Task 3: Review Cloud Logs

Objective

Analyze cloud logs to understand user activities and service operations.

Activities Performed

- Opened AWS event history.
- Reviewed visible AWS CloudTrail management events.
- Examined event names, timestamps, usernames, event sources, resource types, and resource names.

Evidence: AWS Event History Review

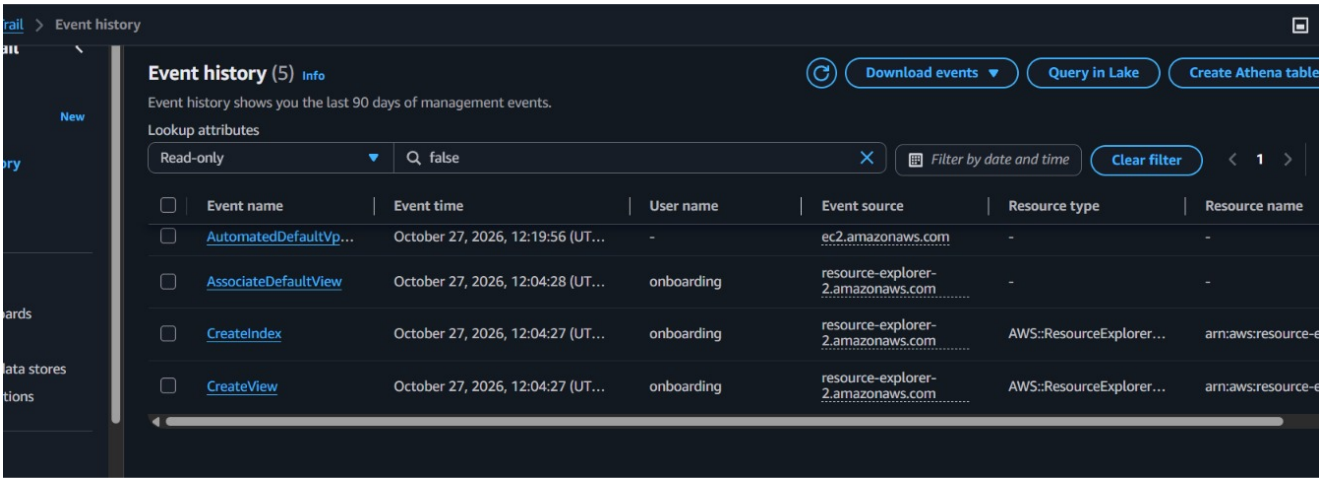


Figure 3. AWS Event history view showing Resource Explorer events and an automated EC2-related event.

Visible Event Summary

Event name	Visible observation
AssociateDefaultView	Resource Explorer default-view association activity.
CreateIndex	A Resource Explorer indexing operation was initiated.
CreateView	A Resource Explorer view was created.
AutomatedDefaultVp…	An automated event is visible; the full event name and details should be opened to confirm its purpose.

Note: The screenshot is a partial view of event history. Confirm full event details, actor identity, request parameters, and outcome before drawing conclusions.

Analysis

- CloudTrail event history supports traceability of AWS management activity.
- Event names alone do not establish whether an action was safe or suspicious; the actor, context, parameters, and expected change should be reviewed.

Recommendation: Enable and retain appropriately protected CloudTrail logs across relevant regions and accounts, centralize logs where possible, set alerts for sensitive administrative actions, and review unusual events regularly.

Task 4: Identify Cloud Misconfigurations

Objective

Identify security weaknesses caused by potentially unsafe cloud configurations, using the visible IAM evidence.

Evidence Reviewed

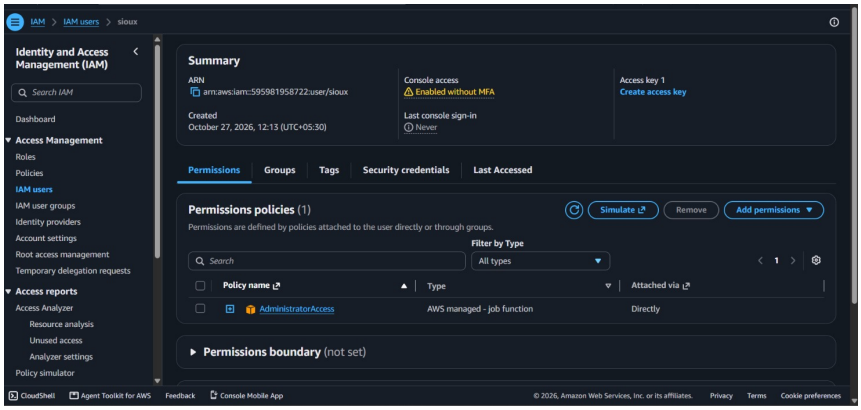


Figure 4. IAM user evidence reused for the configuration review.

Misconfigurations and Risks

Finding	Risk / rationale	Severity
AdministratorAccess attached directly to a user	Broad permissions increase the impact if the account is compromised and make access governance harder.	**High**
Console access without visible MFA	A stolen password may be sufficient to access the console, depending on other controls.	**High**
Direct policy assignment	Can make privilege reviews and consistent access management more difficult; direct attachment is not automatically a vulnerability by itself.	Medium

Severity is a preliminary assessment based on the screenshots and should be validated against business context and compensating controls.

Recommendations

- Enable MFA for all privileged users and enforce stronger sign-in controls.
- Replace broad administrator access with least-privilege policies wherever operationally possible.
- Use IAM groups for job-based user permissions and IAM roles for workloads or temporary access.
- Review access regularly, remove unused credentials, and monitor privileged actions.

Task 5: Create Cloud Security Assessment Report

Executive Summary

A cloud security assessment was conducted using the supplied AWS IAM configuration, AWS management-event history, and security-monitoring evidence. The visible evidence highlights concerns involving privileged access and MFA, along with repeated registry-related events that require validation. The available screenshots do not prove that a compromise has occurred.

Key Findings

- **High risk:** Administrator-level permissions are attached directly to the reviewed IAM user.
- **High risk:** The IAM summary indicates console access without MFA.
- **Investigate:** Repeated registry-related events appear in the supplied security-monitoring view.
- **Improvement area:** Strengthen access governance, centralized logging, and alerting for sensitive activity.

Risk Rating Summary

Category	Risk level	Basis
Identity security	**High**	Privileged console account shown without MFA.
Access management	**High**	AdministratorAccess directly attached to the user.
Logging and monitoring	Medium	Event history is visible; retention, centralization, and alert coverage need verification.
Configuration management	Medium	Registry-related activity needs correlation with approved changes and endpoint evidence.

Recommended Action Plan

Immediate

- Enable MFA for all privileged console users.
- Review the administrator account and confirm that the access level is necessary.
- Investigate the registry events on the affected host and correlate them with authentication and endpoint logs.

Short term

- Implement least-privilege access and remove unnecessary permissions.
- Create alerts for high-impact IAM changes and unusual administrative activity.
- Perform documented periodic access reviews and validate logging coverage.

Long term

- Adopt Zero Trust principles appropriate to the environment.
- Schedule recurring security assessments and configuration reviews.
- Automate compliance checks and alerting, with human validation of findings.

Conclusion

The environment provides useful visibility through IAM configuration pages, CloudTrail event history, and security-monitoring records. The most urgent improvements are MFA enforcement and reducing excessive privileges. Registry-related events should be investigated using corroborating evidence before classifying them as malicious.

Overall Security Rating: MEDIUM RISK (REQUIRES REMEDIATION)